

DS7006 2526 Quantitative Data Analysis

CW2 Final

by

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Abstract

This project examines geographic differences in COVID-19 death rates across 296 local authorities in England by using an ecological dataset which combines deaths caused by Covid-19 with census based indicators. The dependent variable is COVID deaths per 1,000 residents. Independent variables represent four themes including age structure, baseline health, dwelling context, and household deprivation composition. Data preparation was done in SQLite by harmonizing local authority codes, joining multiple tables using LA_code and then standardizing variables as rates per 1,000 residents or per 1,000 households. Exploratory Data Analysis was conducted to assess distributions, outliers, and normality. Bivariate association testing was done using Spearman correlation. Partial correlations were computed to assess adjusted associations and then PCA was applied to summarize correlated indicators into latent dimensions. Multiple regression modelling was used to estimate joint associations while controlling multicollinearity using VIF-based model reduction. Bivariate results show that the strongest association is between mortality and deprivation particularly dep2_per_1000_households ($\rho = 0.422$, $p = 1.355e-14$), poor health also show significant association with mortality ($\rho = 0.401$, $p = 1.002e-12$). The final regression model retained age_75plus_per_1000, dep2_per_1000_households, and unshared_dwellings_per_1000, explaining 21.85% of variance in deaths per 1,000 ($R^2 = 0.2185$; adjusted $R^2 = 0.2105$) with low VIF values. Relative importance analysis indicates that dep2_per_1000_households contributes the largest share of explained variance (approximately 82%). A PCA-score regression robustness check identified the deprivation-health dimension as the dominant component associated with mortality. Overall, deprivation emerges as the strongest correlate of COVID-19 mortality in this dataset, while age distribution and housing context also provide additional explanatory insights, though causal interpretations are limited by the ecological design.

1 Introduction

Coronavirus disease (COVID-19) is an infectious respiratory illness caused by the SARS-CoV-2 virus. This virus spreads mainly through respiratory particles, which are released when an infected person breathes, speaks, coughs or sneezes and the risk increases in crowded and poorly ventilated environments (World Health Organization, 2025). In England, COVID-19 produced substantial mortality with clear geographical variation across different places and population groups. This makes local authority level analysis very useful for understanding unequal impacts (Office for National Statistics, 2020).

A consistent finding during the pandemic is that older age is strongly associated with severe disease and death, because clinical vulnerability increases with age and long term health conditions are more common in older people (World Health Organization, 2025). However, outcomes were not shaped by biology alone. Government evidence reviews and population statistics clearly indicate that social and structural conditions influenced both exposure to infection and the likelihood of severe outcomes. Socio economic deprivation, baseline health disadvantage and housing circumstances have been repeatedly linked to higher COVID-19 risk and worse results (Public Health England, 2020). National statistics also show a pronounced deprivation gradient in mortality, with higher COVID-19 death rates observed in more deprived areas (Office for National Statistics, 2020). Peer-reviewed evidence further supports deprivation as a strong predictor of excess mortality and it shows associations between overcrowding and higher death rate in England (Parkes et al., 2023).

Understanding these relationships at the local authority level remains important for planning of public health, reducing inequalities and improving preparedness for future outbreaks. While ecological analysis cannot prove the cause at individual level, it can identify areas that are most strongly associated with higher mortality and support targeted policy responses.

This study investigates how COVID-19 mortality varies with different socio economic and demographic factors across English local authorities. The dependent variable is COVID deaths per 1,000 residents, and independent variables represent age structure, health status, dwelling context, and deprivation composition. The objective of this study is to identify which indicators and which broader dimensions derived from them show the strongest statistical association with COVID-19 mortality across local authorities in England.

2 Literature Review

COVID-19 death rates varied widely across local authorities in England. The pandemic did not create new inequalities. Instead, it combined with existing social and health disadvantages and often made them worse through “inequality amplification.” (Bambra et al., 2020; McGowan and Bambra, 2022). Although ecological analysis cannot demonstrate the causation at individual level, Local authority level modelling helps identify area-based risk patterns and it supports targeted prevention and preparedness.

Age structure and concentration of clinical risk

Age is one of the leading predictor of severe COVID-19 outcomes. Different linked English health record studies show sharp increase in mortality with age and suggests that local authorities with more older population would be expected to have higher death rates (Williamson et al., 2020). This aligns with the QCOVID risk prediction model, which identifies age as a leading predictor of COVID-19 hospitalization and death (Clift et al., 2020). However, geographic inequalities persist after accounting for age, indicating that age structure alone does not explain local variation and likely interacts with wider social and health conditions (Welsh et al., 2022).

Baseline health disadvantage and self-rated health

Baseline health also shapes COVID-19 mortality. OpenSAFELY studies linked to death registrations show that many predictors of COVID-19 death overlap with predictors of non-COVID death, reflecting broader chronic disease and health inequality patterns (Bhaskaran et al., 2021). When area-level datasets don’t include detailed data on pre-existing conditions, broader population health indicators can be used as substitutes. Self-rated health is commonly used because it captures both diagnosed and undiagnosed illness and is a strong predictor of future mortality (DeSalvo et al., 2006; Bamia et al., 2017). This supports using LA level proportions of good/very good, fair, and bad/very bad health as indicators of underlying vulnerability.

Deprivation as a structural driver

Deprivation is consistently considered as a structural driver of unequal COVID-19 outcomes as it influences both exposure and vulnerability through higher risk jobs, less capacity to self isolate, poorer underlying health, and limited resources. (Bambra et al., 2020; McGowan and Bambra,

2022). OpenSAFELY findings reveal that COVID-19 mortality in England is strongly patterned by deprivation (Williamson et al., 2020). Welsh et al. (2022) similarly report that more deprived local authorities saw earlier and more rapid increases in mortality during the first wave, suggesting deprivation may shape how the epidemic unfolds as well as the overall death toll. Decomposition analyses also underline the role of deprivation by distinguishing between transmission related and vulnerability related pathways that contribute to the deprivation gap (Albani et al., 2022). Evidence from later stages of the pandemic suggests these inequalities continued over time, even if the most important spatial scales changed (Griffith et al., 2022).

Dwelling context and shared-living exposure

As SARS-CoV-2 spreads efficiently indoors, household conditions are an important context for infection spread. A systematic review and meta-analysis found high secondary attack rates within households that shows that the home is a major location for transmission (Madewell et al., 2020).. In England, housing constraints have been linked to mortality. During the first wave, areas with more overcrowded housing experienced higher excess mortality (Davies et al., 2021). While this project uses shared versus unshared dwelling indicators rather than direct overcrowding, qualitative work suggests that shared living can reduce privacy and increase close contact during lockdowns (Blanc and Scanlon, 2022).

Summary and link to research questions

Overall, the literature supports analyzing local authority level mortality using a combined framework: age structure and baseline health capture likelihood of severe outcomes (Clift et al., 2020; Bhaskaran et al., 2021), while deprivation and dwelling context capture structural and environmental influences on exposure and transmission (Bambra et al., 2020; Madewell et al., 2020; Davies et al., 2021; Albani et al., 2022). Because these factors are correlated, multivariable modelling and dimension reduction can help separate overlapping influences and identify broader latent dimensions (Griffith et al., 2022).

3 Research Questions

The study addresses the following research questions.

1. How strongly is the proportion of older residents, especially those aged 65–74 and 75 plus are linked to COVID-19 deaths per 1,000 across English local authorities??
2. Is poorer population health, shown by higher levels of fair health and bad/very bad self-reported health, linked to higher COVID-19 deaths per 1,000 across local authorities?
3. Are housing conditions, measured using the share of shared and unshared dwellings, linked to COVID-19 deaths per 1,000 across English local authorities?
4. Is household deprivation composition linked to COVID-19 mortality, meaning areas with more households in higher deprivation groups (especially dep3+ and dep4) have higher COVID-19 deaths per 1,000?
5. When age, health, housing, and deprivation are included in the same model, which theme and which specific indicators show the strongest independent link with COVID-19 deaths per 1,000 across English local authorities?

These research questions guide the subsequent methods and statistical analysis, including exploratory analysis, correlation assessment, and multivariable modelling.

4 Methodology

4.1 Research design and unit of analysis

This project applies a quantitative and cross sectional ecological design to examine how socio-economic and demographic characteristics are associated with COVID-19 mortality across England. The unit of analysis is the English Local Authority (LA), and the final analytical dataset contains $n = 296$ local authorities. The primary objective is to identify which socio-economic variables show the strongest relationship with COVID-19 mortality and which remain significant once considered together in multivariate modelling.

4.2 Data sources and data preparation

The dependent variable dataset (COVID-19 deaths by local authority) was supplied via Moodle. Independent variables were derived from official open data sources at local authority geography, aligned to the themes of age, baseline health, dwelling context, and household deprivation composition. Data were handled as multiple CSV files and imported into SQLite for preparation and joining.

The preparation process had four stages. First, the COVID-19 deaths dataset required harmonization of local authority codes due to boundary changes and reorganizations. SQL updates were applied so that merged or renamed authorities were represented using consistent LA_code values. After harmonization, deaths were aggregated by LA_code to produce a single total deaths value per local authority. This step produced the expected 296-row analytical base.

Second, independent variable tables were imported into SQLite and structured so that each table contained a common key column (LA_code) and a relevant denominator column (Total residents for population-based indicators, and Total households for household deprivation indicators). For the age-structure indicators, NOMIS age bands were recoded and merged into four groups (0–15, 16–49, 50–64, and 65–74) before being converted to rates per 1,000 residents using the total population denominator.

Third, tables were joined using LEFT JOIN operations on LA_code to preserve the full set of local authorities in the COVID dataset while attaching indicators from each theme.

Fourth, standardization was applied to ensure comparability across local authorities. COVID deaths were standardized as deaths per 1,000 residents. Age and health indicators were converted to rates per 1,000 residents using the total population denominator. Dwelling indicators were also converted per 1,000 residents. Deprivation composition indicators were standardized per 1,000 households using total households as the denominator. This standardization was essential because raw counts are strongly influenced by population size, and rates allow meaningful comparison across local authorities with different population and household totals.

4.3 Techniques

All analysis was conducted in R. Packages were used for visualization, correlation, partial correlation, factor adequacy diagnostics, PCA, regression modelling, and multicollinearity checks.

4.3.1 Exploratory Data Analysis pipeline

Exploratory Data Analysis assessed distribution shape, outliers, and normality. Histograms with density curves, boxplots, and Q-Q plots were used for visual diagnostics. Normality was formally tested using Shapiro-Wilk and Kolmogorov-Smirnov tests for a representative subset of key variables.

4.3.2 Correlation analysis

Bivariate association testing used Spearman rank correlations to quantify monotonic relationships between the dependent variable and candidate predictors in a way that is less sensitive to non-normality and outliers. A full Spearman correlation matrix was also computed to assess dependence among predictors and to anticipate multicollinearity risks.

4.3.3 Partial correlation

Partial correlation was used to estimate the association between `deaths_per_1000` and each candidate predictor while controlling for a set of theoretically central confounders representing age, baseline health disadvantage, and deprivation.

Control variables used in partial correlation

age_75plus_per_1000

bad_or_very_bad_health_per_1000

dep2_per_1000_households

Partial correlations were computed on complete cases and used to identify predictors with associations that persisted after adjustment for the controls.

4.3.4 KMO and MSA

KMO and MSA diagnostics were used to assess sampling adequacy for PCA. PCA with varimax rotation was then applied to summarize correlated indicators into broader latent dimensions. Reference categories from compositional blocks were removed prior to PCA to avoid perfect multicollinearity.

4.4 Principal Component Analysis

PCA was applied to reduce dimensionality and summarize correlated indicators into interpretable components. Prior to PCA, reference categories from compositional blocks were removed to avoid perfect multicollinearity and singular correlation matrices. Specifically, age_0_15_per_1000, good_or_very_good_health_per_1000, and dep0_per_1000_households were removed from the PCA input set, leaving 12 variables.

4.4.1 Interpretation of Components

Components were extracted from the Pearson correlation matrix and rotated using varimax rotation to improve interpretability and produce uncorrelated component scores. Four components were retained to achieve a balance between explained variance and interpretability, explaining 0.92 of the cumulative variance. Component interpretation was based on rotated loadings, assigning substantive labels according to variables with the largest absolute loadings.

Component scores were computed for use as alternative predictors in a robust regression model. The resulting rotated component scores were orthogonal.

4.5 Variable selection for regression

Predictor selection was guided by:

- 1) EDA distributional checks and normality diagnostics
- 2) Spearman correlation patterns with deaths_per_1000
- 3) partial correlations controlling for age, bad health, and deprivation
- 4) multicollinearity diagnostics

Given that several predictors are compositional and highly inter correlated, multicollinearity was explicitly evaluated using variance inflation factors.

Reference categories removed to avoid perfect multicollinearity

age_16_49_per_1000

good_or_very_good_health_per_1000

dep0_per_1000_households

4.6 Multiple regression modelling and tuning

Multiple linear regression was applied to model deaths per 1,000 as a function of selected predictors. Model building began with a full model to assess joint effects and diagnose multicollinearity using VIF. Reduced models were then estimated to retain theoretical coverage while improving interpretability and collinearity properties. Stepwise selection using BIC was used as a confirmation check.

4.6.1 Relative importance

Relative importance analysis using the lmg metric was computed for the final model to quantify each predictor's contribution to explained variance.

4.6.2 PCA score regression robustness check

A robustness regression was estimated using PCA component scores as predictors to assess whether the broader latent dimensions derived from the indicator set were associated with deaths_per_1000.

5 Implementation

5.1 Data import and dataset verification

The analysis was implemented in R using the final joined and normalized dataset Final_Covid_Normalized.csv. After import, the dataset structure was checked to confirm that variables were correctly typed, the unit of analysis was consistent across indicators, and the dataset met expected integrity conditions for local authority level analysis.

Main commands

```
read.csv(), str(), nrow(), duplicated()
```

Table 1 Dataset verification checks (n, missingness, duplicates, and variable removal)

```
> nrow(Covid.LA)
[1] 296

> # Missing values per column
> apply(Covid.LA, MARGIN = 2, FUN = function(x) sum(is.na(x)))
      LA_code      LA_name
      0          0
total_covid_deaths deaths_per_1000
      0          0
age_0_15_per_1000  age_16_49_per_1000
      0          0
age_50_64_per_1000 age_65_74_per_1000
      0          0
age_75plus_per_1000 good_or_very_good_health_per_1000
      0          0
fair_health_per_1000 bad_or_very_bad_health_per_1000
      0          0
shared_dwelling_per_1000 unshared_dwelling_per_1000
      0          0
dep0_per_1000_households dep1_per_1000_households
      0          0
dep2_per_1000_households dep3_per_1000_households
      0          0
dep4_per_1000_households dep3plus_per_1000_households
      0          0

> # Duplicate rows
> sum(duplicated(Covid.LA))
[1] 0

> summary(Covid.LA[[DV]])
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000  1.653   2.058   2.072  2.498   4.074
>
```

Interpretation

The dataset contained 296 local authorities. There were no missing values across variables and no duplicate records. A derived deprivation indicator `dep3plus_per_1000_households` was removed prior to analysis because it duplicates information contained in `dep3_per_1000_households` and `dep4_per_1000_households`.

The dependent variable was defined as `deaths_per_1000`. Summary statistics indicated a mean of 2.072 deaths per 1,000 residents, with values ranging from 0.000 to 4.074.

5.2 Exploratory Data Analysis and distribution diagnostics

Exploratory Data Analysis was used to evaluate distributional form, identify outliers, and inform the selection of association measures. Diagnostic plots and normality tests were produced for a small set of representative variables capturing the outcome and key predictors across themes: `deaths_per_1000`, `age_75plus_per_1000`, `bad_or_very_bad_health_per_1000`, `dep2_per_1000_households`, and `shared_dwelling_per_1000`.

Main commands

Histogram with density: `hist(x, probability = TRUE)` and `lines(density(x))`

Boxplot: `boxplot(x)`

Q-Q plot: `qqnorm(x)` and `qqline(x)`

Normality tests: `shapiro.test(x)` and `ks.test(x, "pnorm", mean(x), sd(x))`

Normality testing and hypotheses arising from EDA

H0: For each variable tested, the sample values come from a normal distribution.

H1: For each variable tested, the sample values do not come from a normal distribution.

Normality was assessed using Shapiro-Wilk and Kolmogorov-Smirnov tests. P-values were interpreted using $\alpha = 0.05$. Where normality was not supported, rank-based methods were preferred for bivariate association testing.

EDA also suggested potential monotonic associations between `deaths_per_1000` and key predictors, particularly deprivation and poor health. These exploratory hypotheses were tested using Spearman rank correlation with H0: $\rho = 0$ and H1: $\rho \neq 0$ at $\alpha = 0.05$.

Table 2 Normality test results for selected indicators

```
> normal.tbl
      variable      n      ks_p      shapiro_p
1      deaths_per_1000 296 9.156613e-01 2.528449e-01
2      age_75plus_per_1000 296 8.673163e-01 9.306823e-01
3 bad_or_very_bad_health_per_1000 296 1.286221e-01 4.455841e-05
4      dep2_per_1000_households 296 6.689923e-01 1.124438e-02
5      shared_dwellings_per_1000 296 1.619605e-18 1.305751e-25

>
```

Interpretation

Formal normality testing indicated that some indicators were not normally distributed, particularly shared dwelling rates, and non-normality was also evident for deprivation and poor health indicators. These diagnostics supported the use of Spearman rank correlation for bivariate association testing, as it is less sensitive to skewness and outliers than Pearson correlation.

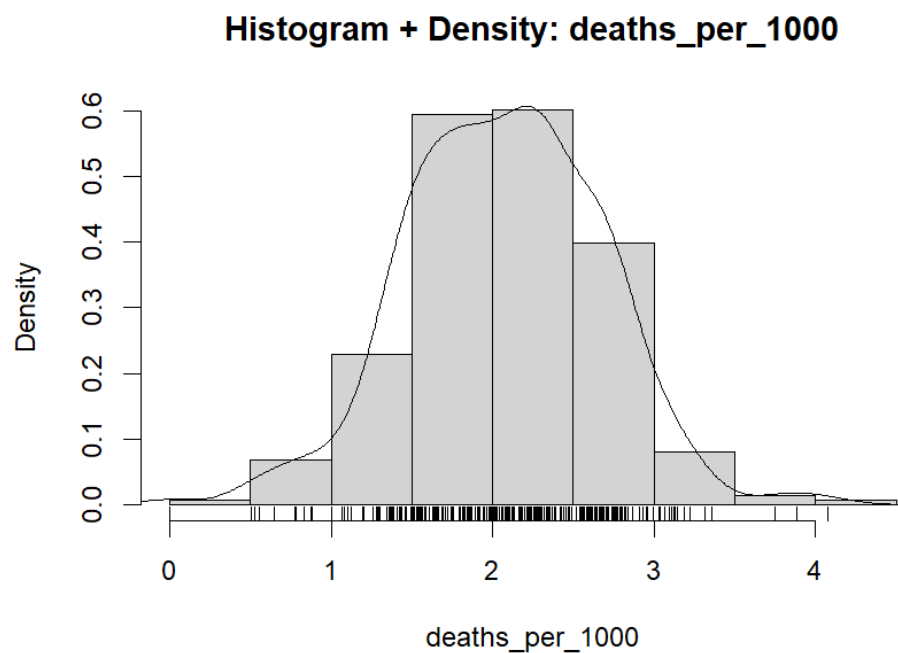


Figure 1 Distribution of COVID deaths per 1,000 residents

Interpretation

The distribution of deaths_per_1000 is unimodal and centred around approximately 2 deaths per 1,000, with a slight right tail. This indicates most local authorities cluster around a similar mortality rate, while a smaller number experienced noticeably higher rates.

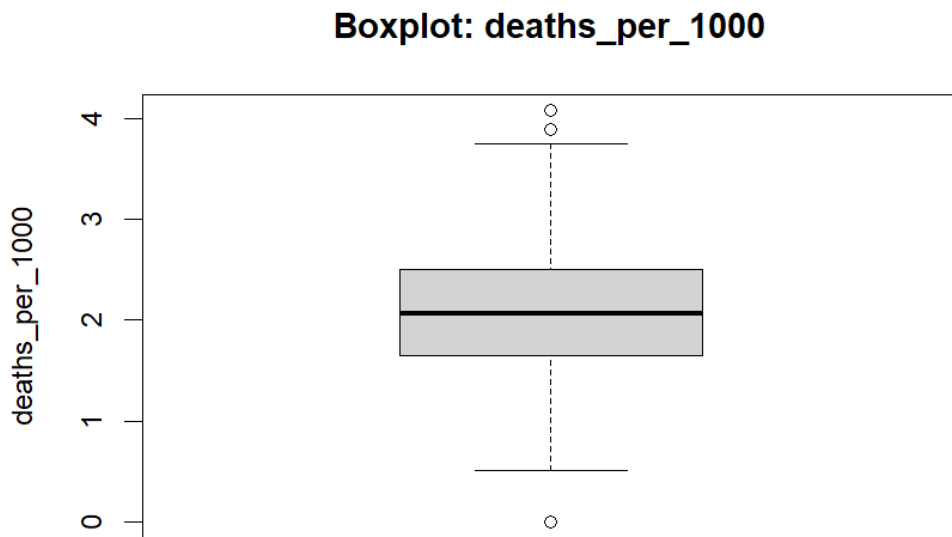


Figure 2 Outlier structure of COVID deaths per 1,000 residents

Interpretation

The boxplot shows moderate spread around the median (just above 2 per 1,000) and highlights a small number of higher-rate outliers above 4 per 1,000, with one very low observation close to 0. This supports checking robustness and diagnostics in later modelling, as a few extreme areas could influence fitted relationships.

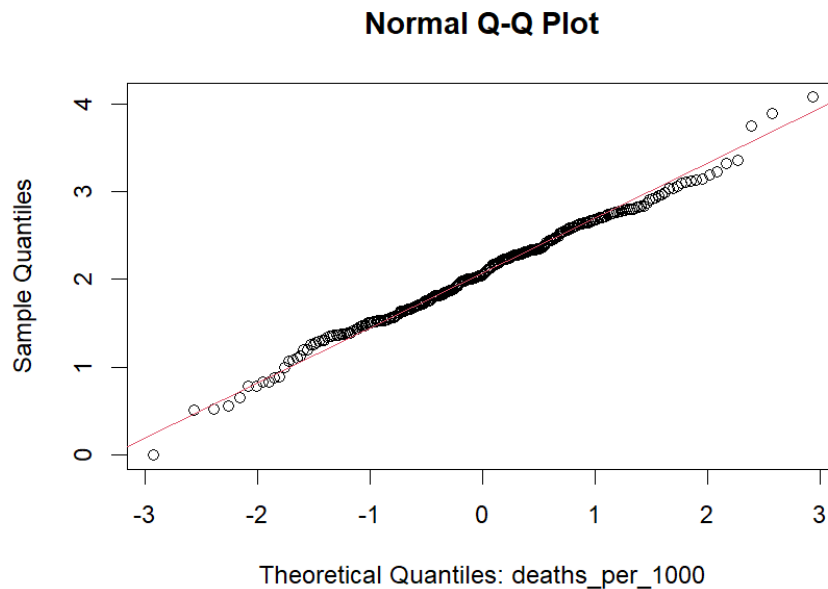


Figure 3 Normal Q–Q plot of COVID deaths per 1,000 residents

Interpretation

The Q–Q plot indicates that `deaths_per_1000` follows an approximately normal pattern in the central range, but deviates in the tails, particularly at the upper end. This pattern is consistent with mild skewness and outliers, supporting the use of robust association measures (for example Spearman correlation) alongside regression diagnostics.

5.3 Bivariate association testing and correlation structure

Bivariate relationships between `deaths_per_1000` and candidate predictors were assessed using Spearman rank correlation. This approach provides a monotonic association measure suitable for non normal distributions and potential outliers. A full Spearman correlation matrix was also computed to understand dependence among predictors and identify multicollinearity risks before regression modelling.

Main commands

```
cor.matrix <- cor(Covid.num, use = "pairwise.complete.obs", method = "spearman")  
cor.test(x, y, method = "spearman")
```

Hypotheses (Spearman correlation)

H0: $\rho = 0$ (no monotonic association between the variables).

H1: ρ is not equal to 0 (a monotonic association exists).

Associations were considered statistically supported when $p < 0.05$.

5.3.1 Correlation ranking output

Table 3 Strongest and weakest Spearman correlations with deaths_per_1000

```
> head(sort(cor.matrix[, DV], decreasing = TRUE), 10)
      deaths_per_1000      total_covid_deaths
1.000000000             0.43741876
dep2_per_1000_households bad_or_very_bad_health_per_1000
0.42232541             0.40071674
fair_health_per_1000      dep3_per_1000_households
0.35418408             0.34631843
dep1_per_1000_households      age_0_15_per_1000
0.27428893             0.21880370
dep4_per_1000_households      age_65_74_per_1000
0.18606310             0.08556754
> tail(sort(cor.matrix[, DV], decreasing = TRUE), 10)
      age_0_15_per_1000      dep4_per_1000_households
0.218803704             0.186063105
      age_65_74_per_1000      age_50_64_per_1000
0.085567540             0.082857896
      age_75plus_per_1000      unshared_dwellings_per_1000
0.040675190             -0.009755827
      shared_dwellings_per_1000      age_16_49_per_1000
-0.055919863             -0.098897357
good_or_very_good_health_per_1000      dep0_per_1000_households
-0.385226521             -0.396513000
```

Interpretation

The strongest bivariate association among the selected predictors was observed between dep2_per_1000_households and deaths_per_1000 ($\rho = 0.422$, $p = 1.355e-14$). Poor health also showed a statistically significant association with mortality ($\rho = 0.401$, $p = 1.002e-12$).

Table 4 Spearman correlation between deprivation and COVID mortality

```
> cor.test(Covid.LA$dep2_per_1000_households, Covid.LA[[DV]], method = "spearman")

Spearman's rank correlation rho

data: Covid.LA$dep2_per_1000_households and Covid.LA[[DV]]
S = 2496906, p-value = 1.355e-14
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho 
0.4223254

>
```

Interpretation

The Spearman test confirms a statistically significant positive association between household deprivation (dep2_per_1000_households) and COVID mortality (deaths_per_1000) ($\rho = 0.422$, $p = 1.355e-14$). This provides formal evidence that local authorities with a higher prevalence of households deprived in two dimensions tended to experience higher COVID death rates.

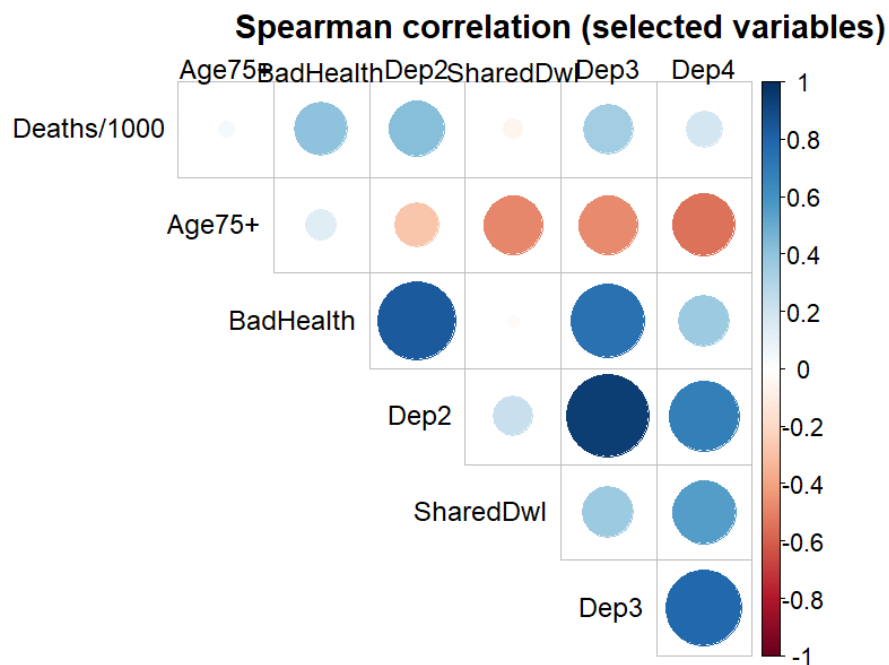


Figure 4 Spearman correlation plot for selected indicators

Interpretation

The correlation plot summarizes the relationship between the outcome and a selected subset of predictors and highlights dependence among predictors. Deaths/1000 shows positive associations with BadHealth, Dep2, and Dep3, while the deprivation indicators are strongly correlated with each other, indicating potential multicollinearity if included together in regression models

5.4 Partial correlation analysis

Partial correlation analysis was used to examine whether associations between deaths_per_1000 and candidate predictors persisted after controlling for key confounding structure. Controls were selected to represent age vulnerability, baseline health disadvantage, and deprivation intensity: age_75plus_per_1000, bad_or_very_bad_health_per_1000, and dep2_per_1000_households.

Main command

```
pcor.test(y, x, controls_dataframe)
```

Key result summary

Partial correlations indicated that several bivariate associations attenuated after adjustment, consistent with overlap between themes. The adjusted analysis supported variable prioritization for regression by focusing on predictors that showed evidence of an independent association beyond the control set.

5.5 KMO, MSA, and Principal Component Analysis implementation

KMO and MSA diagnostics were applied to evaluate the suitability of PCA. PCA was then used to summarize correlated indicators into broader latent dimensions. To avoid perfect multicollinearity, reference categories from compositional blocks were removed prior to PCA input construction.

Main commands

```
KMO(R)
```

```
eigen(R)
```

```
principal(R, nfactors = 4, rotate = "varimax", scores = FALSE)
```

```
principal(Covid.pca, nfactors = 4, rotate = "varimax", scores = TRUE)
```

Table 5 KMO and MSA results for PCA suitability

```
> KMO(R)
Kaiser-Meyer-Olkin factor adequacy
Call: KMO(r = R)
Overall MSA = 0.75
MSA for each item =
      age_16_49_per_1000      age_50_64_per_1000
              0.77              0.77
      age_65_74_per_1000      age_75plus_per_1000
              0.86              0.80
      fair_health_per_1000 bad_or_very_bad_health_per_1000
              0.82              0.69
      shared_dwellings_per_1000      unshared_dwellings_per_1000
              0.87              0.48
      dep1_per_1000_households      dep2_per_1000_households
              0.60              0.70
      dep3_per_1000_households      dep4_per_1000_households
              0.73              0.82

>
```

Interpretation

Sampling adequacy was acceptable for PCA (overall MSA = 0.75). Item level MSA values were inspected, with `unshared_dwellings_per_1000` showing a lower MSA (0.48), which suggests weaker shared variance with the remaining indicators .

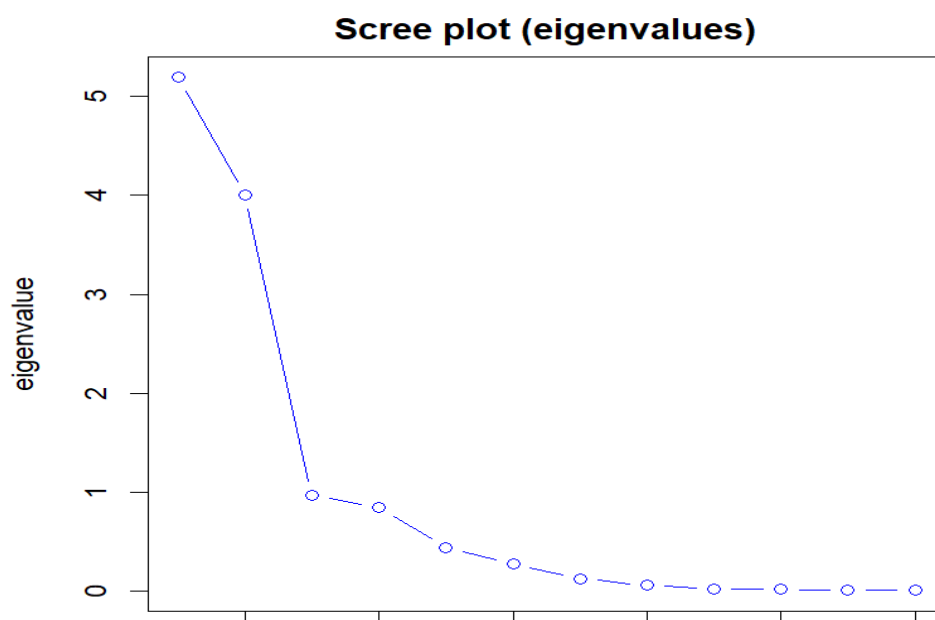


Figure 5 Scree plot of PCA eigenvalues

Interpretation

The scree plot shows a steep decline in eigenvalues for the first components, followed by a clear flattening, indicating that a small number of components capture most of the shared variation in the indicators.

Table 6 Varimax-rotated PCA loadings and variance explained

```
> fit <- principal(R, nfactors = 4, rotate = "varimax", scores = FALSE)
> fit
Principal Components Analysis
Call: principal(r = R, nfactors = 4, rotate = "varimax", scores = FALSE)
Standardized loadings (pattern matrix) based upon correlation matrix
```

	RC1	RC2	RC4	RC3	h2	u2	com
age_16_49_per_1000	-0.95	-0.07	0.18	0.00	0.95	0.054	1.1
age_50_64_per_1000	0.89	-0.01	-0.19	0.23	0.89	0.110	1.2
age_65_74_per_1000	0.95	0.11	-0.15	0.20	0.98	0.021	1.2
age_75plus_per_1000	0.95	-0.01	-0.08	0.15	0.94	0.063	1.1
fair_health_per_1000	0.52	0.80	-0.11	0.11	0.94	0.060	1.8
bad_or_very_bad_health_per_1000	0.13	0.92	-0.12	0.24	0.92	0.076	1.2
shared_dwellings_per_1000	-0.28	-0.01	0.91	-0.04	0.92	0.084	1.2
unshared_dwellings_per_1000	0.28	0.07	-0.02	0.93	0.96	0.043	1.2
dep1_per_1000_households	0.32	0.78	0.20	-0.19	0.79	0.213	1.6
dep2_per_1000_households	-0.23	0.96	0.06	-0.05	0.99	0.015	1.1
dep3_per_1000_households	-0.51	0.83	0.06	0.08	0.96	0.040	1.7
dep4_per_1000_households	-0.59	0.47	0.45	0.16	0.79	0.207	3.0

	RC1	RC2	RC4	RC3
SS loadings	4.72	3.95	1.21	1.13
Proportion Var	0.39	0.33	0.10	0.09
Cumulative Var	0.39	0.72	0.82	0.92
Proportion Explained	0.43	0.36	0.11	0.10
Cumulative Proportion	0.43	0.79	0.90	1.00


```
Mean item complexity = 1.4
Test of the hypothesis that 4 components are sufficient.

The root mean square of the residuals (RMSR) is 0.04

Fit based upon off diagonal values = 0.99

>
```

Interpretation

The varimax-rotated solution provides interpretable component loadings and summarizes explained variance. In this solution, four components jointly explain 0.92 of total variance, indicating strong shared structure across age composition, health, deprivation, and dwelling indicators.

5.6 Multiple regression modelling and multicollinearity control

Multiple linear regression was used to model `deaths_per_1000` as a function of selected predictors. Model building began with a full specification to evaluate joint effects and diagnose multicollinearity. Reduced models were then estimated to preserve theoretical coverage while improving interpretability and collinearity properties. Multicollinearity was assessed using VIF.

5.6.1 Full model estimation and multicollinearity diagnosis

Table 7 Full Regression Model

```
> summary(model.full)

Call:
lm(formula = form.full, data = Covid.reg)

Residuals:
    Min       1Q   Median       3Q      Max
-1.71775 -0.27383 -0.00555  0.28946  1.71625

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      1.5053682   1.3438455    1.120  0.26358
age_0_15_per_1000  0.0058188   0.0024480    2.377  0.01812 *
age_50_64_per_1000  0.0096904   0.0038661    2.507  0.01275 *
age_65_74_per_1000 -0.0109425   0.0074412   -1.471  0.14253
age_75plus_per_1000  0.0147406   0.0049956    2.951  0.00344 **
fair_health_per_1000 -0.0093881   0.0070743   -1.327  0.18556
bad_or_very_bad_health_per_1000  0.0043004   0.0146213    0.294  0.76888
shared_dwellings_per_1000  0.1387917   0.2346893    0.591  0.55473
unshared_dwellings_per_1000 -0.0002799   0.0011833   -0.237  0.81321
dep1_per_1000_households -0.0159689   0.0051264   -3.115  0.00203 **
dep2_per_1000_households  0.0363473   0.0082730    4.393 1.58e-05 ***
dep3_per_1000_households -0.0364602   0.0150802   -2.418  0.01625 *
dep4_per_1000_households  0.0253248   0.0518982    0.488  0.62595
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5046 on 283 degrees of freedom
Multiple R-squared:  0.3293,    Adjusted R-squared:  0.3008
F-statistic: 11.58 on 12 and 283 DF,  p-value: < 2.2e-16

>
```

Table 8 Full model VIF

```
> vif(model.full)
      age_0_15_per_1000      age_50_64_per_1000
           3.185547           8.425042
      age_65_74_per_1000      age_75plus_per_1000
           39.281794           18.940360
      fair_health_per_1000 bad_or_very_bad_health_per_1000
           21.173198           39.627460
      shared_dwellings_per_1000      unshared_dwellings_per_1000
           1.441219           3.280759
      dep1_per_1000_households      dep2_per_1000_households
           9.610592           60.035246
      dep3_per_1000_households      dep4_per_1000_households
           43.680170           4.443387

>
```

Interpretation

The full model was estimated to evaluate joint effects across age, health, dwelling, and deprivation indicators. VIF values indicate severe multicollinearity among several compositional predictors, which can inflate standard errors and destabilize coefficient estimates. This supported moving to a reduced specification with clearer interpretation and acceptable collinearity.

5.6.2 Reduced final model estimation

Table 9 Model finalA (Final regression model)

```
> summary(model.finalA)

Call:
lm(formula = deaths_per_1000 ~ age_75plus_per_1000 + dep2_per_1000_households +
    unshared_dwellings_per_1000, data = Covid.reg)

Residuals:
    Min       1Q   Median       3Q      Max
-1.6721 -0.3033 -0.0103  0.3659  1.5949

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      1.3047859   0.3550886   3.675 0.000283 ***
age_75plus_per_1000  0.0055018   0.0013775   3.994 8.22e-05 ***
dep2_per_1000_households  0.0100802   0.0011747   8.581 5.65e-16 ***
unshared_dwellings_per_1000 -0.0025072   0.0007597  -3.300 0.001085 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5362 on 292 degrees of freedom
Multiple R-squared:  0.2185, Adjusted R-squared:  0.2105
F-statistic: 27.21 on 3 and 292 DF, p-value: 1.51e-15

>
```

Table 10 Model finalA (VIF)

```
> vif(model.finalA)
      age_75plus_per_1000      dep2_per_1000_households
              1.275243              1.071882
unshared_dwellings_per_1000
              1.197384

>
```

Interpretation

A reduced final model was selected with low multicollinearity and clear interpretation:

$\text{deaths_per_1000} \sim \text{age_75plus_per_1000} + \text{dep2_per_1000_households} + \text{unshared_dwellings_per_1000}$

This final model explained 21.85 percent of outcome variance (R squared = 0.2185; adjusted R squared = 0.2105). All predictors were statistically significant and VIF values were close to 1, indicating no multicollinearity concerns.

5.6.3 Model selection confirmation

Table 11 Stepwise model selection output (BIC) for confirmation

```
> summary(model.final.step)

Call:
lm(formula = deaths_per_1000 ~ dep2_per_1000_households + age_75plus_per_1000 +
    unshared_dwellings_per_1000, data = Covid.reg)

Residuals:
    Min       1Q   Median       3Q      Max
-1.6721 -0.3033 -0.0103  0.3659  1.5949

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      1.3047859   0.3550886   3.675 0.000283 ***
dep2_per_1000_households  0.0100802   0.0011747   8.581 5.65e-16 ***
age_75plus_per_1000      0.0055018   0.0013775   3.994 8.22e-05 ***
unshared_dwellings_per_1000 -0.0025072   0.0007597  -3.300 0.001085 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5362 on 292 degrees of freedom
Multiple R-squared:  0.2185,    Adjusted R-squared:  0.2105
F-statistic: 27.21 on 3 and 292 DF,  p-value: 1.51e-15

>
```

Interpretation

Forward stepwise selection using BIC selected the same set of predictors, confirming stability. Relative importance analysis showed that `dep2_per_1000_households` contributed the largest share of explained variance in the final model (approximately 82 percent), with smaller contributions from `age_75plus_per_1000` and `unshared_dwellings_per_1000`.

5.6.4 Explained variance contribution

Table 12 Relative importance of predictors in the final regression model

```
> # Relative importance
> calc.relimp(model.finalA, type = c("lmg"), rela = TRUE)
Response variable: deaths_per_1000
Total response variance: 0.3641042
Analysis based on 296 observations

3 Regressors:
age_75plus_per_1000 dep2_per_1000_households unshared_dwellings_per_1000
Proportion of variance explained by model: 21.85%
Metrics are normalized to sum to 100% (rela=TRUE).

Relative importance metrics:

                                lmg
age_75plus_per_1000             0.09233819
dep2_per_1000_households        0.82277757
unshared_dwellings_per_1000    0.08488424

Average coefficients for different model sizes:

                                1X              2Xs              3Xs
age_75plus_per_1000            0.001030093    0.003066507    0.005501826
dep2_per_1000_households        0.008954900    0.009341313    0.010080240
unshared_dwellings_per_1000    -0.001481885   -0.001662513   -0.002507175

>
```

Interpretation

Relative importance analysis decomposes the explained variance (R^2) across predictors. The results indicate that `dep2_per_1000_households` account for the largest share of explained variance (approximately 82%), while `Age75+` and `unshared dwellings` contribute smaller shares, supporting deprivation as the dominant statistical predictor within this model.

5.7 PCA score regression robustness check

A robustness regression was implemented using PCA component scores, allowing the outcome to be modelled using orthogonal dimensions rather than correlated original indicators.

Main commands

```
lm(deaths_per_1000 ~ RC1 + RC2 + RC3 + RC4, data = Covid.pca.reg)
stepwise(model.pca, direction = "forward")
```

Table 13 PCA score regression outputs(robustness check)

```
> summary(model.pca)

Call:
lm(formula = deaths_per_1000 ~ RC1 + RC2 + RC3 + RC4, data = Covid.pca.reg)

Residuals:
    Min       1Q   Median       3Q      Max
-1.71843 -0.31286 -0.00848  0.36572  1.68672

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.07200    0.03188  65.002 < 2e-16 ***
RC1          0.04264    0.03193   1.335  0.1828
RC2          0.24411    0.03193   7.645 3.05e-13 ***
RC3         -0.07603    0.03193  -2.381  0.0179 *
RC4         -0.01525    0.03193  -0.478  0.6332
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5484 on 291 degrees of freedom
Multiple R-squared:  0.1852, Adjusted R-squared:  0.174
F-statistic: 16.53 on 4 and 291 DF, p-value: 3.206e-12

>
> summary(model.pca.step)

Call:
lm(formula = deaths_per_1000 ~ RC2, data = Covid.pca.reg)

Residuals:
    Min       1Q   Median       3Q      Max
-1.87306 -0.32867 -0.01808  0.39204  1.80116

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.07200    0.03213  64.491 < 2e-16 ***
RC2          0.24411    0.03218   7.585 4.4e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5528 on 294 degrees of freedom
Multiple R-squared:  0.1637, Adjusted R-squared:  0.1608
F-statistic: 57.53 on 1 and 294 DF, p-value: 4.395e-13
>
```

Interpretation

The PCA-score regression models the outcome using orthogonal component scores, avoiding multicollinearity among original indicators. In the full PCA-score model, RC2 is highly

significant, and forward selection retains RC2 alone, indicating that the dominant underlying dimension captured by RC2 explains a substantial share of variation in COVID deaths per 1,000 across local authorities.

6 Discussion

6.1 Significant and most important variables

Across all techniques, deprivation is the strongest and most consistent correlate of COVID-19 mortality. Spearman correlations show `dep2_per_1000_households` has the largest bivariate association with `deaths_per_1000`, and it remains the dominant predictor in the final multivariable regression. Relative importance results indicate approximately 82% of explained variance in the final model is attributable to `dep2_per_1000_households`, far exceeding other retained predictors.

Age structure matters after adjustment. While `age_75plus_per_1000` shows only a weak bivariate relationship with mortality, it becomes statistically significant and positive in the final model once deprivation and dwelling context are controlled. This pattern is consistent with confounding or suppression at LA level, where age structure overlaps with other area characteristics and can obscure the adjusted association.

Dwelling context adds independent information. `unshared_dwellings_per_1000` is significant and negatively associated with mortality in the final model, consistent with the idea that less shared living may reduce indoor transmission risk, though it may also partially proxy wider socio-economic conditions.

6.2 Associations, trends, and multicollinearity

The results show a clear deprivation gradient: higher deprivation composition aligns with higher mortality, whereas low deprivation (`dep0`) and good health indicators show negative associations. Health disadvantage aligns with mortality in bivariate testing, but its independent contribution reduces once deprivation is included, suggesting substantial shared variance between deprivation and health.

Predictors are highly interdependent. Correlation patterns and VIF diagnostics indicate strong overlap, especially among deprivation and age composition measures. PCA reinforces this structure: four components explain 0.92 cumulative variance, implying the variables reflect a smaller number of underlying dimensions. PCA-score regression identifies a dominant component (RC2) associated with mortality, consistent with a deprivation–health disadvantage dimension driving much of the variation.

6.3 Link to research questions

The findings answer the research questions in a combined framework. Age contributes to higher mortality once modelled jointly (vulnerability mechanism). Baseline health disadvantage is positively related to mortality in bivariate analyses, but its effect is partly absorbed by deprivation in multivariable models. Deprivation is the clearest determinant, with dep2_per_1000_households most influential across bivariate and regression-based approaches. Dwelling context is weaker overall, but unshared dwellings remain significant and predicts lower mortality after adjustment. Joint modelling therefore highlights deprivation as dominant, and PCA suggests the key latent structure linked to mortality is a deprivation–health dimension.

6.4 Causal interpretation and limits

The associations are consistent with plausible mechanisms, deprivation shaping exposure (work, isolation constraints, housing) and susceptibility via linked health disadvantage; age increasing clinical vulnerability; and dwelling structure affecting household transmission. However, inference must remain cautious because the design is ecological and unmeasured confounding is likely. Local authorities differ in factors not modelled here (e.g., ethnicity, density, care home prevalence, healthcare access, and wave timing). Results should therefore be read as area-profile associations, not individual-level causal effects

7 Conclusion

This project examined local authority variation in COVID-19 mortality in England using indicators representing age structure, baseline health, dwelling context, and household deprivation. The evidence consistently indicates that deprivation is the strongest statistical correlation of COVID deaths per 1,000 residents. Bivariate testing showed a moderate positive association between mortality and dep2_per_1000_households ($\rho = 0.422$), with poor health also significantly associated ($\rho = 0.401$). Due to strong dependence among compositional indicators, the full regression model exhibited severe multicollinearity, so a reduced model was selected for stable interpretation.

The final regression model retained age_75plus_per_1000, dep2_per_1000_households, and unshared_dwellings_per_1000, explaining 21.85 percent of variance in deaths per 1,000 with low VIF values. Relative importance analysis indicates dep2_per_1000_households accounts for the largest share of explained variance (approximately 82 percent). A PCA-score robustness regression similarly identifies the deprivation-health disadvantage dimension as the primary component associated with mortality. Overall, the analysis suggests that local authorities characterized by higher deprivation, older age structure, and less favorable dwelling profiles experienced higher COVID-19 mortality, while interpretation remains constrained by the ecological nature of the data and potential unmeasured confounding.

8 References

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9 Appendix A

1. Final_Covid_Normalized.csv

10 Appendix A1

Sql queries

```
BEGIN TRANSACTION;

-- Shepway
UPDATE COVID_19_deaths
SET LA_code = 'E07000112',
    LA_name = 'Folkestone and Hythe'
WHERE LA_name = 'Shepway';

-- Somerset West and Taunton
UPDATE COVID_19_deaths
SET LA_code = 'E0000246',
    LA_name = 'Somerset West and Taunton'
WHERE LA_code IN ('E07000190','E07000191');

-- West Suffolk
UPDATE COVID_19_deaths
SET LA_code = 'E07000245',
    LA_name = 'West Suffolk'
WHERE LA_code IN ('E07000201','E07000204');

-- East Suffolk
UPDATE COVID_19_deaths
SET LA_code = 'E07000244',
    LA_name = 'East Suffolk'
WHERE LA_code IN ('E07000205','E07000206');

-- East Staffordshire
UPDATE COVID_19_deaths
SET LA_code = 'E07000193'
WHERE LA_name = 'East Staffordshire';

-- East Riding of Yorkshire
UPDATE COVID_19_deaths
SET LA_code = 'E06000011'
WHERE LA_name = 'East Riding of Yorkshire';

-- East Northamptonshire
UPDATE COVID_19_deaths
```

```

SET LA_code = 'E07000152'
WHERE LA_name = 'East Northamptonshire';

-- Dorset
UPDATE COVID_19_deaths
SET LA_code = 'E06000059',
    LA_name = 'Dorset'
WHERE LA_code IN
('E07000049', 'E07000050', 'E07000051', 'E07000052', 'E07000053');

-- Bournemouth, Christchurch and Poole
UPDATE COVID_19_deaths
SET LA_code = 'E06000058',
    LA_name = 'Bournemouth, Christchurch and Poole'
WHERE LA_code IN ('E06000028', 'E07000048', 'E06000029');

-- Buckinghamshire
UPDATE COVID_19_deaths
SET LA_code = 'E06000060',
    LA_name = 'Buckinghamshire'
WHERE LA_code IN ('E07000004', 'E07000005', 'E07000006', 'E07000007');

-- Northampton
UPDATE COVID_19_deaths
SET LA_code = 'E07000154'
WHERE LA_name = 'Northampton';

-- West Northamptonshire
UPDATE COVID_19_deaths
SET LA_code = 'E06000062',
    LA_name = 'West Northamptonshire'
WHERE LA_code IN ('E07000151', 'E07000154', 'E07000155');

-- North Northamptonshire
UPDATE COVID_19_deaths
SET LA_code = 'E06000061',
    LA_name = 'North Northamptonshire'
WHERE LA_code IN ('E07000150', 'E07000152', 'E07000153', 'E07000156');

-- Somerset West and Taunton
UPDATE COVID_19_deaths
SET LA_code = 'E07000246',
    LA_name = 'Somerset West and Taunton'
WHERE LA_code = 'E0000246';

-- Somerset
UPDATE COVID_19_deaths
SET LA_code = 'E06000066',
    LA_name = 'Somerset'
WHERE LA_code IN ('E07000187', 'E07000188', 'E07000189', 'E07000246');

-- North Yorkshire
UPDATE COVID_19_deaths
SET LA_code = 'E06000065',
    LA_name = 'North Yorkshire'

```

```
WHERE LA_code IN
('E07000163','E07000164','E07000165','E07000166','E07000167','E07000168','E0
7000169');

-- Westmorland and Furness
UPDATE COVID_19_deaths
SET LA_code = 'E06000064',
    LA_name = 'Westmorland and Furness'
WHERE LA_code IN ('E07000027','E07000030','E07000031');

-- Cumbria
UPDATE COVID_19_deaths
SET LA_code = 'E06000063',
    LA_name = 'Cumbria'
WHERE LA_code IN ('E07000026','E07000028','E07000029');

COMMIT;

DROP TABLE IF EXISTS Updated_Dataset;

CREATE TABLE Updated_Dataset AS
SELECT
    d.LA_code,
    MAX(d.LA_name) AS LA_name,
    SUM(d.Total) AS Total
FROM COVID_19_deaths d
INNER JOIN Age a
    ON d.LA_code = a.LA_code
GROUP BY d.LA_code;

SELECT COUNT(*) AS n_rows
FROM Updated_Dataset;
```